

Mandatory Assignment 3

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- **Deadline: October 09, 2023 by 23:59.**
 - The submission is in the form of a .pdf file which should be submitted through LearnIT.
 - We suggest that you either use LaTeX or scan hand-written notes to create .pdf.
 - You are encouraged to use the reference material while attempting the problems.
 - The mandatory assignment should be submitted individually. This does not mean that you cannot work in groups to solve these, but you must write up your solution in your own words and submit that.
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1. Let X be a random variable taking values $\{0, 1, 2, 3, \dots\}$. Which of the following are valid PMFs?
 - (a) Let $q \in (0, 1)$ and $p_X(x) = q$ if $x = 1$, $p_X(x) = 1 - q$ if $x = 0$ and $p_X(x) = 0$ for all other values of x .
 - (b) $p_X(x) = \frac{x-1}{5}$ for $x = 0, 1, 2, 3, 4$ and $p_X(x) = 0$ for all other values of x .
 - (c) $p_X(x) = \left(\frac{1}{2}\right)^x$ for $x = 0, 1, 2, \dots, 10$ and $p_X(x) = 0$ for all other values of x .
 - (d) $p_X(x) = \left(\frac{1}{2}\right)^x$ for $x = 0, 1, 2, \dots$
 - (e) Let $q \in (0, 1)$ and $p_X(k) = \binom{n}{k} \cdot q^k \cdot (1 - q)^{n-k}$ for $0 \leq k \leq n$ and $p_X(k) = 0$ for the rest of the values of k .
2. (Exercise 3.44 parts (a), (b)) For x, y binary digits (0 or 1), let $x \oplus y$ be 0 if $x = y$ and 1 if $x \neq y$. (This operation is called exclusive or which is abbreviated as XOR. It is also called addition mod 2.)
 - (a) Let $p \in (0, 1)$, let $X \sim \text{Bern}(p)$, and $Y \sim \text{Bern}(1/2)$, independently. What is the distribution of $X \oplus Y$?
 - (b) With notation as in (a), is $X \oplus Y$ independent of X ? Is $X \oplus Y$ independent of Y ? Be sure to consider both the case $p = 1/2$ and the case $p \neq 1/2$.

1. Let X be a random variable taking values $\{0, 1, 2, 3, \dots\}$. Which of the following are valid PMFs?

(a) Let $q \in (0, 1)$ and $p_X(x) = q$ if $x = 1$, $p_X(x) = 1 - q$ if $x = 0$ and $p_X(x) = 0$ for all other values of x .

$$f_X(X = \{0, 1, 2, \dots, n\})$$

Conditions to have valid PMF: 1.) Sum = 1
2.) Non neg. numbers

$$\begin{aligned} a) \quad q &\in (0, 1) & p_X(x) &= q \text{ if } (x) = 1 \\ & & p_X(x) &= 1 - q \text{ if } (x) = 0 \\ & & p_X(x) &= 0 \text{ if else} \end{aligned}$$

$$1.) \text{ Sum} = ? \rightarrow q + 1 - q + 0 = 1 \Rightarrow 1/2$$

$$2.) \text{ Non neg} \rightarrow 1, 0 \rightarrow \text{non } \geq 0 \wedge q \in (0, 1) \text{ so } q \geq 0 \Rightarrow \text{Non neg. } 2/2$$

2/2 $\rightarrow$ valid PMF

b) (b) $p_X(x) = \frac{x-1}{5}$ for $x = 0, 1, 2, 3, 4$ and $p_X(x) = 0$ for all other values of x .

$$\begin{aligned} p_X(x) &= \frac{x-1}{5} \text{ for } x \in \{0, 1, 2, 3, 4\} \\ p_X(x) &= 0 \text{ for } x < 0 \wedge x > 4 \end{aligned}$$

if I plug $\{0\}$ for x , then $\frac{-1}{5}$. Its neg. value and valid PMF can't be negative number
so NOT VALID PMF

c) (c) $p_X(x) = \left(\frac{1}{2}\right)^x$ for $x = 0, 1, 2, \dots, 10$ and $p_X(x) = 0$ for all other values of x .

$$\begin{aligned} p_X(x) &= \left(\frac{1}{2}\right)^x \text{ for } x = \{0, 1, 2, \dots, 10\} \\ p_X(x) &= 0 \text{ for others} \end{aligned}$$

$$1.) \text{ Sum must be equal to zero} \rightarrow \text{if } x=0, \text{ then } \left(\frac{1}{2}\right)^0 = 1$$

$\rightarrow$ so if I add $x \in \{1, \dots, 10\}$, sum will be $> 1 \Rightarrow$ Not valid PMF

d)

$$(d) \quad p_X(x) = \left(\frac{1}{2}\right)^x \text{ for } x = 0, 1, 2, \dots$$

Here its also NOT VALID PMF, because if $x=0$ then $\left(\frac{1}{2}\right)^0 = 1$. then SUM will be > 1

- (e) Let $q \in (0, 1)$ and $p_X(k) = \binom{n}{k} \cdot q^k \cdot (1-q)^{n-k}$ for $0 \leq k \leq n$ and $p_X(k) = 0$ for the rest of the values of k .

$$p_X(k) = \binom{n}{k} \cdot q^k \cdot (1-q)^{n-k} \quad \text{for } k \geq 0 \wedge k \leq n$$

So $k \in \langle 0, n \rangle$

$$p_X(k) = 0 \quad \text{for } k < 0$$

1.) So the sum =

$$\sum_{k=0}^n \binom{n}{k} q^k \cdot (1-q)^{n-k}$$

↳ k will be looping and adding 1 to k each iteration, and summing the values until $k == n$

Since I already know that's binomial distribution I can say that
(this was said to be enough in my lecture) ↳ its valid PMF

2. (Exercise 3.44 parts (a), (b)) For x, y binary digits (0 or 1), let $x \oplus y$ be 0 if $x = y$ and 1 if $x \neq y$. (This operation is called exclusive or which is abbreviated as XOR. It is also called addition mod 2.)

- (a) Let $p \in (0, 1)$, let $X \sim \text{Bern}(p)$, and $Y \sim \text{Bern}(1/2)$, independently. What is the distribution of $X \oplus Y$?

$$X = \{0, 1\} \quad Y = \{0, 1\}$$

$$X \oplus Y = 0 \quad \text{is} \quad X = Y$$

$$X \oplus Y = 1 \quad \text{is} \quad X \neq Y$$

$$p = p, 1$$

$$X \sim \text{Bern}(p) \quad \text{ind.} \\ Y \sim \text{Bern}\left(\frac{1}{2}\right)$$

$$X \oplus Y = \text{DISTRIBUTION?}$$

$$\text{so } P(X=1) = p \quad \Rightarrow P(X=0) = 1-p \\ P(Y=1) = \frac{1}{2} \quad \Rightarrow P(Y=0) = \frac{1}{2}$$

$$P(X=1) + P(X=0) = 1$$

$$P(X=0) + P(Y=0) = 1$$

Case: $X=0 \wedge Y=0$

$$P(X=0 \wedge Y=0) = P(X=0) \cdot P(Y=0) = (1-p) \cdot \frac{1}{2} = \frac{1-p}{2}$$

$$X=Y \Rightarrow \underline{X \oplus Y = 0} \quad \text{for } X=0 \wedge Y=0$$

Case: $X=1 \wedge Y=1$

$$P(X=1 \wedge Y=1) = P(X=1) \cdot P(Y=1) = \frac{p}{2}$$

$$X=Y \Rightarrow \underline{X \oplus Y = 0} \quad \text{for } X=1 \wedge Y=1$$

Case: $X=0 \wedge Y=1$

$$P(X=0 \wedge Y=1) = P(X=0) \cdot P(Y=1) = \frac{1-p}{2}$$

$$X \neq Y \Rightarrow \underline{X \oplus Y = 1}$$

Case $X=1 \wedge Y=0$

$$P(X=1 \wedge Y=0) = P(X=1) \cdot P(Y=0) = \frac{p}{2}$$

$$X \neq Y \Rightarrow \underline{X \oplus Y = 1}$$

Prob. for $X \oplus Y = 1$

$$\begin{aligned} &\Downarrow \\ &\frac{1-p}{2} + \frac{p}{2} \\ &= \frac{1-p+p}{2} = \frac{1}{2} \quad \text{for } X \oplus Y = 1 \end{aligned}$$

Prob. for $X \oplus Y = 0$

$$\begin{aligned} &\frac{1-p}{2} + \frac{p}{2} \\ &= \frac{1}{2} \quad \text{for } X \oplus Y = 0 \end{aligned}$$

(b) With notation as in (a), is $X \oplus Y$ independent of X ? Is $X \oplus Y$ independent of Y ? Be sure to consider both the case $p = 1/2$ and the case $p \neq 1/2$.

considering both cases $p = \frac{1}{2}$ and $p \neq \frac{1}{2}$ is not necessary because I calculated this above: $\frac{1-p+p}{2} = \frac{1}{2} \quad (X \oplus Y \sim \text{Bern}(\frac{1}{2}))$

$\Downarrow$ $-p+p$ $\rightarrow$ So value of p is not important

a) is $X \oplus Y$ independent of X ?

if $(X \oplus Y = 0, X = 0)$ will be equal to $(X \oplus Y = 0, X = 1)$, then X is not influencing

case: $\downarrow$
 $X=0 \wedge Y=0$

case: $\downarrow$
 $X=1, Y=1$

Since changing value of $x \in \{0, 1\}$ won't influence $X \oplus Y$, so: $X \oplus Y$ is independent of X

b) if $(X \oplus Y = 0, Y = 0) = (X \oplus Y = 0, Y = 1)$, then Y won't influence the $X \oplus Y \Rightarrow$

$\downarrow$
 $X=0, Y=0$

$=$ $\downarrow$
 $X=1, Y=1$

$\Rightarrow$ $X \oplus Y$ is independent of